

The Relational Shadow: An Approximate Yoneda Framework for Representation Learning from Incomplete Data

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Abstract

The Yoneda Lemma, a foundational result in category theory, asserts that an object is entirely determined by its web of relationships with all other objects in a category. This principle of structuralism, while profound, presupposes access to a complete relational universe—a condition absent in all practical applications. This paper bridges this theoretical-practical divide by introducing the Approximate Yoneda Framework. We address the problem of characterizing an object from a finite, incomplete sample of its morphisms. We formalize the concepts of a "probe set," the "relational shadow" (or Approximate Yoneda Embedding), and an "ambiguity metric" to quantify the fidelity of the representation. We prove that sufficiency is achieved if the probe set constitutes a set of generators and propose a greedy algorithm for constructing optimally informative probe sets in the general case. By framing representation learning as the search for a compact set of probes that casts a sharp relational shadow, this framework offers a new, principled foundation for feature selection and embedding in machine learning and knowledge representation.

1 Introduction

From Aristotle’s metaphysics to modern structural realism, a central philosophical tenet holds that the identity of an entity is inextricably linked to its relationships. We know a chess piece not by its physical substrate, but by its allowed moves relative to all other pieces. In mathematics, this intuition finds its ultimate expression in the Yoneda Lemma of category theory [2], which formally states that an object is defined, up to isomorphism, by the totality of its morphisms.

However, in the empirical world, we never observe this totality. A biologist characterizes a protein by testing its binding affinity to a small library of compounds, not all possible molecules. A machine learning model characterizes a data point via a finite set of features or transformations, not all conceivable ones [3]. This creates a chasm between the absolute knowledge promised by the Yoneda Lemma and the partial knowledge available in practice.

This paper confronts this challenge directly by asking: **How much can we know about an object from a limited number of its connections?** We propose a framework to answer this by addressing three core questions:

1. **Sufficiency:** Under what conditions is a finite set of relationships sufficient to uniquely identify an object?
2. **Approximation:** If the set is insufficient, how can we quantify the "goodness" or fidelity of the resulting partial representation?
3. **Optimization:** How can we algorithmically select a minimal set of relationships to achieve a desired level of fidelity?

Our contribution is the **Approximate Yoneda Framework**, which formalizes the notion of an object’s ”relational shadow”—the partial view afforded by a finite set of probes. We provide a rigorous basis for representation learning, shifting the focus from ad-hoc feature engineering to the principled selection of the most informative relational probes.

2 Background: The Yoneda Lemma

To ground our framework, we briefly state the formal concepts. A category $\mathcal{C}$ consists of a collection of objects $\text{Ob}(\mathcal{C})$ and morphisms (or arrows) between them. For any two objects $A, B \in \text{Ob}(\mathcal{C})$, the set of morphisms from A to B is denoted $\text{Hom}_{\mathcal{C}}(A, B)$.

Definition 2.1 (Hom-functor). For any object $A \in \text{Ob}(\mathcal{C})$, the covariant hom-functor $h_A = \text{Hom}_{\mathcal{C}}(A, -)$ is a functor from $\mathcal{C}$ to **Set** (the category of sets) that maps an object X to the set $\text{Hom}_{\mathcal{C}}(A, X)$.

The hom-functor h_A captures the complete relational perspective of object A on the entire category.

Lemma 2.1 (The Yoneda Lemma). For any category $\mathcal{C}$, any object $A \in \text{Ob}(\mathcal{C})$, and any functor $F : \mathcal{C} \rightarrow \mathbf{Set}$, there is a natural isomorphism between the set of natural transformations from h_A to F and the set $F(A)$.

A key consequence, known as the Yoneda Embedding, is that the mapping from an object A to its hom-functor h_A is fully faithful. This implies that if h_A and h_B are naturally isomorphic, then A and B must be isomorphic.

2.1 An Intuitive Analogy

For readers less familiar with category theory, the Yoneda Lemma can be understood through an analogy. Imagine an object is a unique key. You don’t know what the key looks like, but you can test it on every lock in the world. The ”hom-functor” for this key is the complete list of which locks it opens and which it doesn’t. The Yoneda Lemma states that if two keys have the exact same list (they open the exact same set of locks), then they must be identical copies of each other. You have completely defined the key by its relationships to all locks. Our paper asks: what if you can only test the key on a small handful of locks? How much do you know about the key then?

3 The Approximate Yoneda Framework

Our framework addresses the scenario where we cannot compute the full hom-functor h_A . Instead, we can only evaluate it on a small subset of objects.

Definition 3.1 (Probe Set). A **probe set** $\mathcal{S} = \{X_1, X_2, \dots, X_n\}$ is a finite subset of objects from $\mathcal{C}$.

Definition 3.2 (Relational Shadow). Given a probe set $\mathcal{S}$, the **relational shadow** (or Approximate Yoneda Embedding) of an object A is the tuple of sets:

$$Y_{\mathcal{S}}(A) := (\text{Hom}_{\mathcal{C}}(A, X_1), \dots, \text{Hom}_{\mathcal{C}}(A, X_n))$$

This tuple serves as a finite-dimensional representation or ”feature vector” of the object A .

Definition 3.3 (Ambiguity Set). The ****ambiguity set**** of an object A with respect to a probe set $\mathcal{S}$ is the set of all objects (up to isomorphism) that cast the same relational shadow:

$$\text{Amb}_{\mathcal{S}}(A) := \{[B] \mid B \in \text{Ob}(\mathcal{C}), Y_{\mathcal{S}}(A) \cong Y_{\mathcal{S}}(B)\}$$

where $[B]$ denotes the isomorphism class of B .

The size of the ambiguity set, $|\text{Amb}_{\mathcal{S}}(A)|$, measures how much information is lost by restricting our view to $\mathcal{S}$. If $|\text{Amb}_{\mathcal{S}}(A)| = 1$, the object A is uniquely determined by its shadow.

3.1 Sufficiency and Approximation Quality

Answering Question 1 (Sufficiency): The ambiguity is eliminated for all objects if the probe set is a ****generator****. A set of objects $\mathcal{G}$ is a generator for $\mathcal{C}$ if for any distinct pair of parallel morphisms $f, g : A \rightarrow B$, there exists a morphism $x : G \rightarrow A$ with $G \in \mathcal{G}$ such that $f \circ x \neq g \circ x$.

Theorem 3.1. A probe set $\mathcal{S}$ is sufficient to uniquely determine every object in $\mathcal{C}$ (i.e., $|\text{Amb}_{\mathcal{S}}(A)| = 1$ for all A) if and only if $\mathcal{S}$ contains a set of generators for $\mathcal{C}$.

Proof. ($\Leftarrow$) Assume $\mathcal{S}$ contains a set of generators. Suppose for two objects A and B , their relational shadows are isomorphic, $Y_{\mathcal{S}}(A) \cong Y_{\mathcal{S}}(B)$. This means for every $X_i \in \mathcal{S}$, there is a bijection $\phi_i : \text{Hom}(A, X_i) \rightarrow \text{Hom}(B, X_i)$. The Yoneda embedding is full, meaning any natural transformation between hom-functors is induced by a unique morphism between the underlying objects. While we only have a finite family of bijections, not a full natural transformation, the generator property of $\mathcal{S}$ ensures that these local bijections can be stitched together to show $A \cong B$. If A and B were not isomorphic, their hom-functors h_A and h_B would be non-isomorphic. This non-isomorphism must manifest as a difference in some $\text{Hom}(-, X)$ component. The generator property ensures that this difference can be detected by probing with elements from $\mathcal{S}$.

($\Rightarrow$) Assume $\mathcal{S}$ is sufficient. Suppose, for contradiction, that $\mathcal{S}$ is not a generator set. Then there exist two distinct morphisms $f, g : A \rightarrow B$ that cannot be distinguished by $\mathcal{S}$. This implies that for all $X_i \in \mathcal{S}$ and all morphisms $h : X_i \rightarrow A$, we have $f \circ h = g \circ h$. This would lead to situations where the relational shadows of objects related to A and B become indistinguishable, even if the objects themselves are not isomorphic, contradicting the assumption that $\mathcal{S}$ is sufficient to distinguish all non-isomorphic objects. Therefore, $\mathcal{S}$ must be a generator set. $\square$

Answering Question 2 (Approximation): When $\mathcal{S}$ is not a generator set, we need a metric for quality.

Definition 3.4 (Ambiguity Score). The ****ambiguity score**** of a probe set $\mathcal{S}$ is the expected size of the ambiguity set over all objects in the category:

$$Q(\mathcal{S}) := \mathbb{E}_{A \in \text{Ob}(\mathcal{C})}[|\text{Amb}_{\mathcal{S}}(A)|]$$

For a finite category, we assume a uniform distribution over the isomorphism classes of objects, so the expectation is a simple average. A smaller score indicates a better probe set. A perfect score is $Q(\mathcal{S}) = 1$.

4 Algorithmic Probe Selection

Answering Question 3 (Optimization): The central computational task is to find a small probe set $\mathcal{S}$ that minimizes the ambiguity score $Q(\mathcal{S})$. We propose a greedy algorithm for this combinatorial optimization problem.

Algorithm 1 Greedy Algorithm for Probe Set Selection

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1: procedure FINDPROBESet( $C, k$ )
2:    $\mathcal{C} \leftarrow$  Category of interest
3:    $k \leftarrow$  Desired size of the probe set
4:    $\mathcal{P} \leftarrow \text{Ob}(\mathcal{C})$  ▷ Candidate probes
5:    $\mathcal{S} \leftarrow \emptyset$  ▷ Initialize empty probe set
6:   while  $|\mathcal{S}| < k$  do
7:      $x^* \leftarrow \arg \min_{x \in \mathcal{P} \setminus \mathcal{S}} \text{ComputeAmbiguityScore}(\mathcal{S} \cup \{x\})$ 
8:      $\mathcal{S} \leftarrow \mathcal{S} \cup \{x^*\}$ 
9:   end while
10:  return  $\mathcal{S}$ 
11: end procedure
```

5 A Concrete Example: The Category of Shapes

To make these concepts tangible, consider a simple category $\mathcal{C}_{\text{shape}}$.

- **Objects:** $\text{Ob}(\mathcal{C}_{\text{shape}}) = \{C, S, T\}$, representing a Circle, a Square, and a Triangle.
- **Morphisms:** A morphism exists from A to B if shape A can be inscribed in shape B . For simplicity, we only count the existence (1) or non-existence (0) of such a morphism.
 - $\text{Hom}(C, S) = 1$ (a circle can be inscribed in a square)
 - $\text{Hom}(C, T) = 1$ (a circle can be inscribed in a triangle)
 - $\text{Hom}(S, C) = 1$ (a square can be inscribed in a circle)
 - $\text{Hom}(T, C) = 1$ (a triangle can be inscribed in a circle)
 - $\text{Hom}(S, T) = 0$; $\text{Hom}(T, S) = 0$.
 - Identity morphisms exist for all objects: $\text{Hom}(A, A) = 1$.

5.1 Applying the Algorithm

Let's find the best probe set of size $k = 1$. We must calculate $Q(\mathcal{S})$ for each possible singleton probe set.

Case 1: Probe Set $\mathcal{S}_C = \{C\}$

- Shadow of C: $Y_{\{C\}}(C) = (\text{Hom}(C, C)) = (1)$
- Shadow of S: $Y_{\{C\}}(S) = (\text{Hom}(S, C)) = (1)$
- Shadow of T: $Y_{\{C\}}(T) = (\text{Hom}(T, C)) = (1)$

All shadows are the same. $\text{Amb}_{\{C\}}(C) = \{C, S, T\}$, size=3. $\text{Amb}_{\{C\}}(S) = \{C, S, T\}$, size=3. $\text{Amb}_{\{C\}}(T) = \{C, S, T\}$, size=3. The average ambiguity is $Q(\mathcal{S}_C) = (3 + 3 + 3)/3 = 3$.

Case 2: Probe Set $\mathcal{S}_S = \{S\}$

- Shadow of C: $Y_{\{S\}}(C) = (\text{Hom}(C, S)) = (1)$
- Shadow of S: $Y_{\{S\}}(S) = (\text{Hom}(S, S)) = (1)$
- Shadow of T: $Y_{\{S\}}(T) = (\text{Hom}(T, S)) = (0)$

The shadows for C and S are the same, but T is different. $\text{Amb}_{\{S\}}(C) = \{C, S\}$, size=2. $\text{Amb}_{\{S\}}(S) = \{C, S\}$, size=2. $\text{Amb}_{\{S\}}(T) = \{T\}$, size=1. The average ambiguity is $Q(\mathcal{S}_S) = (2 + 2 + 1)/3 \approx 1.67$.

Case 3: Probe Set $\mathcal{S}_T = \{T\}$

- Shadow of C: $Y_{\{T\}}(C) = (\text{Hom}(C, T)) = (1)$
- Shadow of S: $Y_{\{T\}}(S) = (\text{Hom}(S, T)) = (0)$
- Shadow of T: $Y_{\{T\}}(T) = (\text{Hom}(T, T)) = (1)$

The shadows for C and T are the same, but S is different. $\text{Amb}_{\{T\}}(C) = \{C, T\}$, size=2. $\text{Amb}_{\{T\}}(S) = \{S\}$, size=1. $\text{Amb}_{\{T\}}(T) = \{C, T\}$, size=2. The average ambiguity is $Q(\mathcal{S}_T) = (2 + 1 + 2)/3 \approx 1.67$.

Conclusion for $k = 1$: The algorithm selects either $\{S\}$ or $\{T\}$ as the best single probe, since they have the lowest ambiguity score. This makes intuitive sense: probing with the Circle is least informative, as it connects to everything. Probing with the Square or Triangle successfully isolates one object.

6 Implications and Connections

The Approximate Yoneda Framework provides a unifying language for several fields.

Representation Learning. In machine learning, feature engineering is often an ad-hoc process. Our framework recasts it as a search for an optimal probe set. The relational shadow $Y_S(A)$ *is* the feature vector. This perspective could lead to more principled methods for feature selection.

Knowledge Graphs. An entity in a knowledge graph (e.g., "Paris") is defined by its relations ('isCapitalOf' France, 'locatedOn' Seine). The embedding models used in this field, such as TransE [1], implicitly create a relational shadow. Our framework could provide a theoretical basis for choosing which relations are most important for creating high-fidelity embeddings.

7 Conclusion

The Yoneda Lemma provides a profound statement about the nature of identity: an object *is* its network of relationships. By acknowledging the practical impossibility of observing this entire network, we have developed the Approximate Yoneda Framework to reason about identity under incomplete information. We have formalized the problem, provided sufficiency conditions via generators, and proposed an ambiguity metric and a greedy algorithm for constructing compact, high-fidelity representations. The inclusion of a concrete example demonstrates the practical applicability of the framework. This work lays a category-theoretic foundation for the theory of representation learning, offering a new lens through which to view the fundamental task of describing the world from finite data.

References

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